

1. Find the sum & product of the Eigenvalues of the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & -6 \end{bmatrix}$

Solution:

Sum of the Eigen values = sum of the main diagonal elements
 $= 2 + 3 + (-6) = -1$

Product of the Eigen value = $|A|$

$$= 2 \begin{vmatrix} 3 & 1 \\ 1 & -6 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 2 & -6 \end{vmatrix} + 2 \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}$$

$$= 2(-18-1) - 1(-6-2) + 2(1-6) = -40$$

Sum = -1 and Product = -40

2. The product of two Eigenvalues of the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ is 16. Find the third

Eigenvalue of A.

Solution:

Given: The product of two Eigen values of A is 16

(i.e) $\lambda_1 \lambda_2 = 16$

By property, Product of Eigen values = $|A|$

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$$\lambda_1 \lambda_2 \lambda_3 = |A|$$

$$16\lambda_3 = \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix}$$

$$= 6(9-1) + 2(-6+2) + 2(2-6) = 32$$

$$\lambda_3 = \frac{32}{16} = 2$$

The third Eigen value is 2.

3. Two Eigenvalues of the matrix $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$ are 3 and 0. What is the third eigenvalue?

What is the product of the eigenvalues of A?

Solution:

Given: If $\lambda_1 = 3$, $\lambda_2 = 0$, and $\lambda_3 = ?$

By property, Sum of the Eigenvalues = sum of the main diagonals.

$$\lambda_1 + \lambda_2 + \lambda_3 = 8 + 7 + 3 = 18$$

$$3 + 0 + \lambda_3 = 18$$

$$\lambda_3 = 18 - 3 = 15$$

By property, Product of the Eigen values = $|A|$

$$(3)(0)(15) = |A|$$

$$|A| = 0$$

The third eigenvalue is 15, The product of the eigenvalues of A is 0.

4. If 3 and 15 are two Eigenvalues of the matrix $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$ then find the third

eigenvalue and hence = $|A|$

Solution:

Given: If $\lambda_1 = 3$, $\lambda_2 = 15$, and $\lambda_3 = ?$

By property, Sum of the Eigenvalues = sum of the main diagonals.

$$\lambda_1 + \lambda_2 + \lambda_3 = 8 + 7 + 3 = 18$$

$$3 + 15 + \lambda_3 = 18$$

$$\lambda_3 = 18 - 18 = 0$$

By property, Product of the Eigenvalues = $|A|$

$$(3)(15)(0) = |A|$$

$$|A| = 0$$

The third eigenvalue is 0, The product of the eigenvalues of A is 0.